CSC1029 OO Programming

Lecture 3 – Procedural Programming Fundamentals (Part 2)

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THE STORY SO FAR - JAVA PROGRAM STRUCTURE

```
public class HelloWorld {

  public static void main(String args[]) {


  }
}
```

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THE STORY SO FAR – PRIMITIVE DATA

```
public class HelloWorld {

  public static void main(String args[]) {
     int i = 5;                // i is of type integer
     double d = 2.5;           // d is of type double
     boolean b = true;         // b is of type boolean
     char ch = ‘a’;            // a is of type character

     System.out.println(“Value of i is “ + i);
  }
}
```

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THE STORY SO FAR – KEYBOARD INPUT

```
import java.util.Scanner;
public class HelloWorld {


     public static void main(String args[]) {
        Scanner input = new Scanner(System.in);
        System.out.print("Enter number: ");
        int number = input.nextInt();
        System.out.println("Number : " + number);
     }

}
```

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THE STORY SO FAR – SELECTION

```
import java.util.Scanner;
public class HelloWorld {

   public static void main(String args[]) {
      Scanner input = new Scanner(System.in);
      System.out.print("Enter number: ");
      int number = input.nextInt();
      if ( number < 100 ) {
         System.out.println("Small");
      }
      else {
         System.out.println("Large");
      }
   }
}
```

WHAT'S TO COME TODAY?

A review of procedural programming concepts:

- non-primitive types (String objects)
- iteration (loops)
- an introduction to program design

Non-Primitive Types : Strings

```
String name = "Michael"
```

- You can obtain the length of a string

```
int n = name.length(); // will return 7
```

- A string of length 0 is an empty string and can be denoted by “”
- You can access a single character of a string

```
char letter = name.charAt(0);
// will return 'M'
```

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Program Structuring: Iteration (while loop) (1)

Pseudocode: ***The 'while' statement***

The loop condition

```
year = 0
interest = 0
balance = 10000
while balance < 20,000
    year = year + 1
    interest = balance * 0.05
    balance = balance + interest
print out the value of year
```

The loop body

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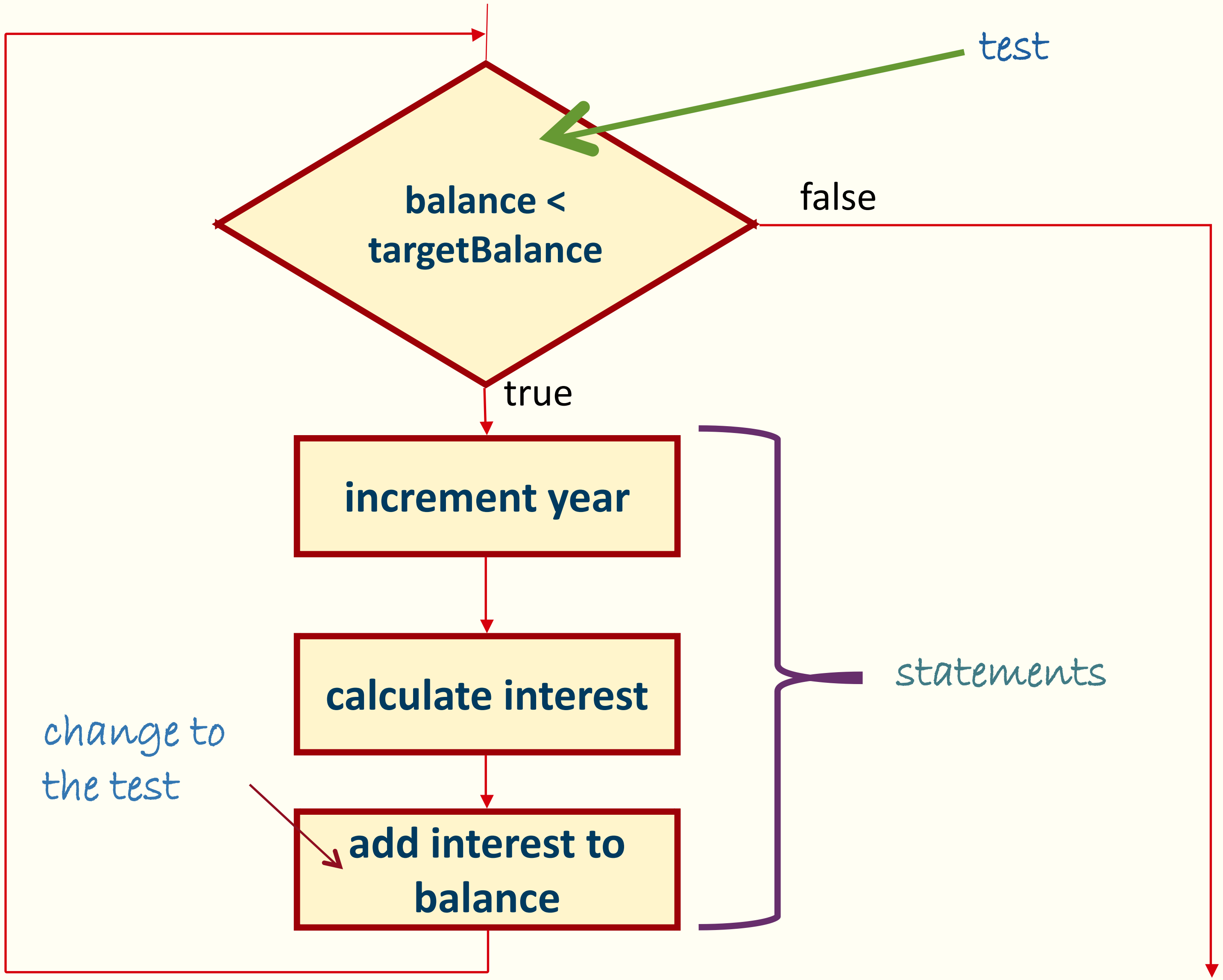
Program Structuring: Iteration (while loop) (2)



Program Structuring: Iteration (while loop) (3)

```
while (condition) {
    statements
}
```

variable is declared outside the loop and updated in the loop

If the condition never becomes false, an infinite loop occurs

```
double balance = 0;

while (balance < targetBalance) {

    year++;
    double interest = balance * RATE / 100;
    balance = balance + interest;

}
```

these statements are executed while the condition is true

this variable is created in each loop iteration

The ‘for’ Loop (1)

```
for (int index=1; index<=5;index++)
{
   System.out.println(“Index:”+index);
}
```

The ‘for’ Loop (2)

Loop Control Variable

```
for (int index=1; index<=5;index++)
{
    System.out.println("Index:"+index);
}
```

The ‘for’ Loop (3)

Loop Control Variable

Boolean Condition

```
for (int index=1; index<=5;index++)
{
   System.out.println("Index:"+index);
}
```

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The ‘for’ Loop (4)

Loop Control Variable

Control Variable Modification

Boolean Condition

```
for (int index=1; index<=5;index++)
{
    System.out.println(“Index:”+index);
}
```

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The ‘for’ Loop (5)

Loop Control Variable **Control Variable Modification**

Boolean Condition

```
for (int index=1; index<=5;index++)
{
    System.out.println("Index:"+index);
}
```

Loop Body

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Program Structuring: Iteration (for loop) (1)

Program Structuring: Iteration (do loop) (2)

- It is sometimes necessary to execute the body of a loop ***at least once*** and then perform the condition test.

```
do {
    statements
}
while (condition);
```

The ‘do .. while‘ Loop (1)

```
int index = 1;
do {
   System.out.println("Index:"+index);
   index++;
} while (index<=5);
```

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The ‘do .. while‘ Loop (2)

```
int index = 1;         Loop Control Variable
do {
    System.out.println("Index:"+index);
    index++;           Control Variable Modification
} while (index<=5);    Boolean Expression
```

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Examples

Loops1.java

Loops2.java

Loops3.java

Case Study: From Design to Implementation (1)

Design and implement a program called “WordWrap” which:

1. asks the user to input a word from the keyboard
2. uses the console to display the word according to the following pattern:

```
Enter a single word: Banana

Banana
a    n
n    a
a    n
n    a
ananaB
```

Using pen/paper:

-Identify what data is required
-Produce an outline design

Case Study: From Design to Implementation (2)

Need to display a message

```
Enter a single word: Banana
```

Need to read a String

```
Banana
a    n
n    a
a    n
n    a
ananaB
```

Display the word as it is

display letters going forward
display letters going backwards

Display the word in reverse

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WHAT'S NEXT?

Program Structure:

- methods
- the role of methods in program structuring
- method parameters and return types

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